Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT | JEE-MAIN | Date: 29-03-2026

Time: 09.00AM TO 12.00PM | GTM-32 | Max. Marks: 300

KEY SHEET

MATHEMATICS

1	**4**	2	**2**	3	**4**	4	**2**	5	**3**
6	**3**	7	**1**	8	**3**	9	**1**	10	**1**
11	**3**	12	**2**	13	**1**	14	**1**	15	**1**
16	**2**	17	**3**	18	**2**	19	**3**	20	**4**
21	**8**	22	**4**	23	**5**	24	**1**	25	**56**

PHYSICS

26	**1**	27	**2**	28	**2**	29	**3**	30	**3**
31	**3**	32	**4**	33	**3**	34	**4**	35	**1**
36	**4**	37	**1**	38	**1**	39	**1**	40	**2**
41	**1**	42	**1**	43	**2**	44	**4**	45	**1**
46	**64**	47	**4**	48	**105**	49	**500**	50	**3**

CHEMISTRY

51	**3**	52	**2**	53	**1**	54	**3**	55	**1**
56	**2**	57	**1**	58	**2**	59	**3**	60	**2**
61	**4**	62	**1**	63	**1**	64	**3**	65	**2**
66	**3**	67	**1**	68	**3**	69	**4**	70	**3**
71	**45**	72	**0**	73	**11**	74	**126**	75	**3**

For More Join: @IITJEE_Advanced

SOLUTION
MATHEMATICS

1. $a_{\min} = \sin^2\theta - \sin\theta + \frac{1}{2}$

$= \sin^2\theta - 2\sin\theta\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 + \frac{1}{2} = \left(\sin\theta - \frac{1}{2}\right)^2 + \frac{1}{4}$

$a_{\min} = \frac{1}{4} \quad \left(\because \sin\theta = \frac{1}{2}\right)$

$b = \lim_{x\to\infty}\left(\sqrt{(x+1)(x+2)} - x\right)$

$\lim_{x\to\infty} \frac{(x+1)(x+2)-x^2}{\sqrt{(x+1)(x+2)}+x} \lim_{x\to\infty} \frac{x\left(3+\frac{2}{x}\right)}{x\left(\sqrt{\left(1+\frac{3}{x}+\frac{2}{x^2}\right)}+1\right)} = \frac{3+0}{1+1} = \frac{3}{2}$

$|2a+b| = \left|2\frac{1}{4} + \frac{3}{2}\right| = \frac{4}{2} = 2$

2. $f(x) = x^3 + e^{x/2} \to f'(a) = 3x^2 + \frac{1}{2}e^{x/2} \quad f'(0) = \frac{1}{2} \quad g(x) = f'(x) \quad$ put $x = f(x)$

$g[f(x)] = f'(fx) = x \quad g'f(x)f'(x) = 1$

$f(x) = 1 \Rightarrow x^2 + e^{x/2} = 1 \Rightarrow x = 0 \quad g'(1)f'(0) = 1 \; g'(1)\frac{1}{2} = 1 \Rightarrow g'(1) = 2$

3. $\int_{\pi/6}^{\pi 3} \frac{1}{\cos^{2/3} x \sin^{4/3} x}dx = \int_{\pi/6}^{\pi/3} \frac{\sec^2 x}{\tan^{4/3} x}dx = \left(-3(\tan)^{-1/3}\right)_{\pi/6}^{\pi/3} = 3^{\frac{7}{6}} - 3^{\frac{5}{6}}$

4. Required Probability $= \frac{666+285-95}{200} = P$ By using addition theorem of probability

$500P = 214$

5. $S_k = \sum_{r=1}^{k} \tan^{-1}\left(\frac{\left(\frac{2}{3}\right)^r}{\left(\frac{2}{3}\right)^{2r}\cdot 2+3}\right) = \sum_{r=1}^{k} \tan^{-1}\left(\frac{\left(\frac{2}{3}\right)^r - \left(\frac{2}{3}\right)^{r+1}}{1+\left(\frac{2}{3}\right)^r\left(\frac{2}{3}\right)^{r+1}}\right)$

$= \sum_{r=1}^{k} \tan^{-1}\left(\tan^{-1}\left(\frac{3}{2}\right)^r - \tan^{-1}\left(\frac{2}{3}\right)^{r+1}\right) \quad S_\infty = \tan^{-1}\left(\frac{2}{3}\right) = \cot^{-1}\left(\frac{3}{2}\right)$

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6. Given $\bar{x}=4$ $n=5$ and $\sigma^2=5.2$ $\sigma^2=\dfrac{\sum(x_i-\bar{x})^2}{n}=5.2$

$$\frac{(x_1-4)^2+(x_2-4)^2+(1-4)^2+(2-4)^2+(6-4)^2}{5}=5.2$$

$$(x_1-4)^2+(x_2-4)^2=9\ldots\ldots(1)$$

$$\bar{x}=4\Rightarrow\frac{x_1+x_2+1+2+6}{5}=4$$

$$x_1+x_2=11\ldots\ldots\ldots(2)$$

from (1) and (2)

$x_1,x_2=4,7$

$x_1x_2=11\Rightarrow x_2=11-x_1$

$(x_1-4)^2+(11-x_1-4)^2=9$

$(x_1-4)^2+(7-x_1)^2=9$

$x_1^2+16-8x_1+49+x_1^2-14x_1=9$

$2x_1^2-22x_1+56=0$ $2x_1^2-11x_1+28=0$

$x_1(x_1-7)-4(x_1-7)=0$ $x_1=4,7$

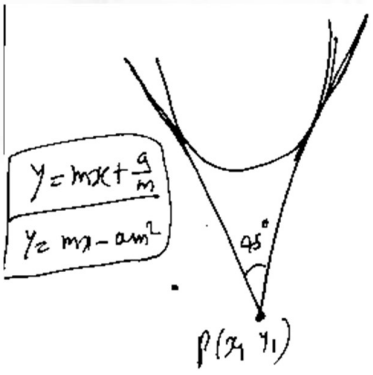
$x_1=4\Rightarrow x_2=11-7=7$

$x_1=7\Rightarrow x_2=11-7=4$ $\therefore x_1,x_2=4,7$

7.

$x^2=y$ $4a=1$

$x^2=4ay$ $a=\dfrac{1}{4}$



Equation of tangent to $x^2=4ay$ is

$y=mx-am^2$ $y=mx-\dfrac{1}{4}m^2$

$4y=4mx-m^2$ $m^2-4mx+4y=0$

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$am^2+bm+c=0 \qquad a=1 \qquad b=-4x \qquad c=4y$

$$m_1+m_2=-\frac{b}{a}=\frac{4x}{x(1)}=4x$$

$$m_1m_2=\frac{c}{a}=\frac{4y}{1}=4y \qquad \tan 45^\circ=\frac{m_1-m_2}{1+m_1m_2}$$

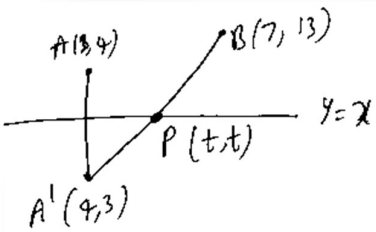
$$1+m_1m_2=\sqrt{(m_1+m_2)^2-4m_1m_2}$$

$$(1+4y)^2=(4x)^2-4(4y)$$

$$1+16y^2+8y=16x^2-16y$$

$$16y^2-16x^2+24y+1=0 \qquad k=24$$

8.



Equation of line BA' is $BA'\ y-3=\frac{10}{3}(x-4)$

$$3y-9=10x-40 \quad 10x-3y-31=0 \quad y=x$$

$$10x-3x-31=0 \quad 7x=31\Rightarrow x=\frac{31}{7} \qquad y=\frac{31}{7}$$

9. Note that

$$\alpha=\cos\left(\frac{8\pi}{11}\right)+\sin\left(\pi-\frac{8\pi}{11}\right)=\cos\left(\frac{8\pi}{11}\right)+\sin\left(\frac{8\pi}{11}\right)$$

$$\therefore \alpha^{11}=\cos(8\pi)+\sin(8\pi)=1$$

Now, $\sum_{k=0}^{10}\alpha^k=\frac{1-\alpha^{11}}{1-\alpha}=0$ $\therefore$ Statement – II is true

We have $\bar{\alpha}=\frac{\alpha\bar{\alpha}}{\alpha}=\frac{1}{\alpha}=\alpha^{10}etc$

Thus, $\mathrm{Re}\left(\alpha+\alpha^2\alpha^3+\alpha^4+\alpha^5\right)$

$$=\frac{1}{2}\left(\alpha+\alpha^2\alpha^3+\alpha^4+\alpha^5+\bar{\alpha}+\bar{\alpha}^2+\bar{\alpha}^1+\bar{\alpha}^4+\bar{\alpha}^5\right)$$

$$=\frac{1}{2}\sum_{k=1}^{10}\alpha^k=\frac{1}{2}(-1)=-\frac{1}{2}$$

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$\therefore$ Statement I is also correct and Statement II is correct explanation for Statement I.
Statement II is true as it is formula for sum of a G.P.

10. $xf(x)=3f^2(x)+2$

Diff w.r.t. x

$f(x)+xf'(x)=3\ \ 2f(x)f'(x)$

$f(x)+xf'(x)=6f(x)f'(x)\quad f(x)=(6f(x)-x)f'(x)$

$$f'(x)=\frac{f(x)}{6f(x)-x}\qquad I=\int\frac{2x^2-12xf(x)+f(x)}{(6f(x)-x)\left(x^2-f(x)\right)^2}dx$$

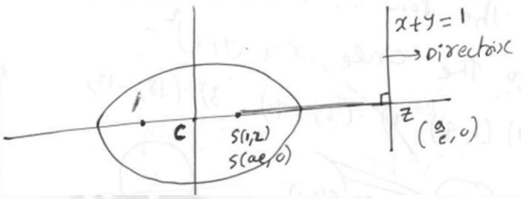
$$=\int\frac{2x(x-6f(x))+f(x)}{(6f(x)-x)\left(x^2-f(x)\right)^2}dx\ =\int\frac{-2x(6f(x)-x)+f(x)}{(6f(x)-x)\left(x^2-f(x)\right)^2}dx$$

$$=\int\frac{\frac{-2x(6f(x)-x)}{6f(x)-x}+\frac{f(x)}{6f(x)-x}}{\left(x^2-f(x)\right)^2}dx=\int\frac{1(2x-f'(x))}{\left(x^2-f(x)\right)^2}dx$$

$$\int\frac{-1}{x^2}dx=\frac{1}{x}+C$$

Put $x^2-f(x)=t\Rightarrow(2x-f'(x))dx=dt=-\int\frac{dt}{t^2}=+\frac{1}{t}+C=\frac{+1}{x^2-f(x)}+C$

11.



$Sz=$ The distance of $S(1,2)$ from the directrix

$x+y=1$ is $\left|\frac{1+2-1}{\sqrt{2}}\right|=\frac{2}{\sqrt{2}}=\sqrt{2}$

$$Sz=\frac{a}{e}-ae=\sqrt{2}\Rightarrow\sqrt{2}a-\frac{a}{\sqrt{2}}=\sqrt{2}\Rightarrow 2a-a=2\Rightarrow a=2$$

$a>b,\ \ b^2=a^2\left(1-e^2\right)$

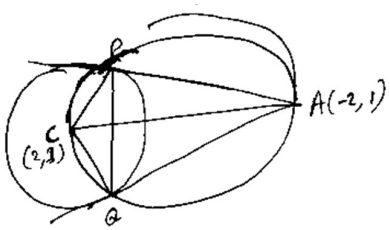
$$b^2=a^2\left(1-e^2\right)=4\left(1-\frac{1}{2}\right)=\frac{4}{2}=2$$

$b^2=2\Rightarrow b=\sqrt{2}$

Length of minor axis $=2b=2\sqrt{2}$

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12.



The equation of circumscribes of circle ΔPAQ

$(x-2)(x+2)+(y-3)(y-1)=0$

$x^2-4+y^2-4y+3=0 \qquad x^2+y^2-4y-1=0$

13. $A)$ If $(x)=(x-1)(x-3)+k(x-2)(x-4)=0$

$(k\in R),(1+k)x^2-(4+6k)x+3+8k=0$

$\Delta=k^2+k+1>0;\forall K\in R$

$B)(x-k)(x-10)=-1\Rightarrow 1\times -1\to k=8\Rightarrow -1\times 1\to k=12$

$C)$ Let $f(x)=\sec x+\cos ecx\Rightarrow f^1(x)=\dfrac{\sin^3 x-\cos^3 x}{\sin^3 x\cos^3 x}\Rightarrow f(x)=0\Rightarrow x=\dfrac{\pi}{4}$

$f(x)\sin$ changes $f(x)=\sqrt{2}+\sqrt{2}=2\sqrt{2}$

$f(x)$ has real roots if $k\in\left(2\sqrt{2},\infty\right)$

$D)\Delta 0\Rightarrow k\in(-\infty,-1)\cup(4,\infty)$

$Sum>0\Rightarrow k>-5\Rightarrow k\in(-5,\infty)$

Product $>0\Rightarrow k>-5\Rightarrow k\in(-5,-1)$

$(1)\cap(2)\cap(3)\Rightarrow k\in(-5,-1)$

14. The number of ways $5c_1 4c_4+5c_2 4c_3+5c_3 4c_2+5c_4 4c_1+5c_5 4c_0=9c_5=126$

15. 3, 7, 11, 15, 19, 23……..407

2, 9, 16, 23, …….709

First common term of both the series is 23 and common difference

$d=l.c.m.\{4,7\}=28$

last term ≤ 407 $a+(n-1)d\le 407$

$23+(n-1)28\le 407$ $28a-5\le 407$ $28n\le 412 \quad n\le\dfrac{412}{28}=14.71$ $n\le 14.71$

Number of terms in common is 14.

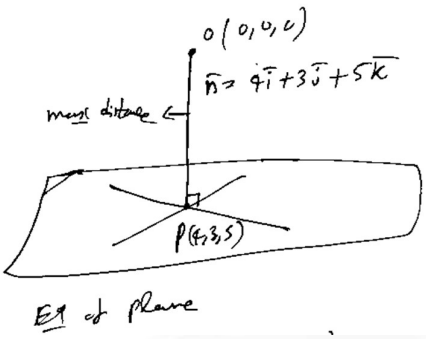
16. $\dfrac{x-1}{3}=\dfrac{y-2}{1}=\dfrac{z-3}{x}=\lambda$

$P=(3\lambda+1,\lambda+2+2\lambda+3)$ lies on (1)

$\dfrac{3\lambda+1-3}{1}=\dfrac{\lambda+2-1}{2}=\dfrac{2\lambda+3-2}{3}$

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$6\lambda-4=\lambda+1\Rightarrow\lambda=1\therefore p=(4,3,5)$



Eq of plane

$4(x-4)+3(y-3)+5(z-5)=0$

$4x+3y+5z-16-9-28=0 \qquad 4x+3y+5z=50$

17. $f(x)=\sin^{-1}\left(\sqrt{x^2+x+1}\right)$

$g(x)=x^2+x+1=x^2+x+\frac{1}{4}+\frac{3}{4}\sin^{-1}x$

$x\in[-1,1] \quad x^2+x+1\le 1 \quad \sqrt{x^2+x+1}\le 1$

$g(x)=\left(x+\frac{1}{2}\right)^2+\frac{3}{4} \ at\ x=-\frac{1}{2}$

$g(x)_{\min}=\frac{3}{4} \qquad g(x)_{\max}=1$

$g(x)\in\left[\frac{3}{4},1\right] \qquad \sqrt{g(x)}=\left[\frac{\sqrt{3}}{2},1\right]$

x^2+x+1

$ax^2+bx+c=\frac{4ac-b^2}{4a}=\frac{4-1}{4}=\frac{3}{4}$

$a=1>0$

$\sin^{-1}\left(\sqrt{g(x)}\right)=\left[\sin^{-1},\left(\frac{\sqrt{3}}{2}\right),\sin^{-1}(1)\right]=\left[\frac{\pi}{3},\frac{\pi}{2}\right]$

18. Since $R=\{(1,2)(2,3)\} \quad A=\{1,2\} \quad B=\{2,3\}$

Now if R_1, is the reflexive relation

Such that $R_1=\{(1,2)(2,3)(1,1)(2,2)(3,3)\}$ has 5 elements

Now if R_2 is both symmetric and reflexive relation

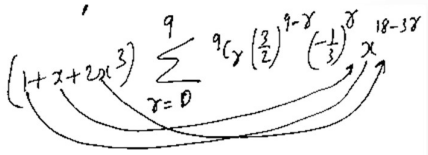
$R_2=\{(1,2)(2,3)(1,1)(2,2)(2,1)(3,2)(3,3)\}$ has 7 elements

Again if R_3 is reflexive, symmetric and transitive all together

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$R_3 = \begin{Bmatrix} (1,2)(2,3)(1,1) \\ (2,2)(2,1)(3,2) \\ (3,3)(1,3)(3,1) \end{Bmatrix}$ has 9 elements starting from 2 elements. Therefore the minimum number of elements to be added is 7

19.



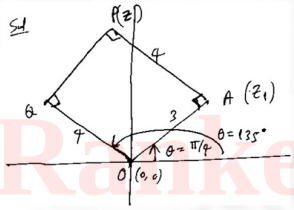
$x^{18-3r} = x^0 \qquad 18-3r=0 \Rightarrow r=\frac{18}{3}=6$

$18-3r+1=0 \qquad 19-3r=0 \Rightarrow r=\frac{19}{3}\notin z^+$

$18-3r+3=0 \qquad \Rightarrow 3r=21 \Rightarrow r=7$

The coefficient of x is ${}^9C_6\left(\frac{3}{2}\right)^3\left(\frac{-1}{3}\right)^6+{}^9C_7\left(\frac{3}{2}\right)^2\left(\frac{-1}{3}\right)^7\times 2=\frac{17}{54}$

20.



$A=(x_1+r\cos\theta, y_1+r\sin\theta)=\left(0+3\cos\frac{\pi}{4}, 0+3\sin\frac{\pi}{4}\right)$

$=\left(\frac{3}{\sqrt{2}},\frac{3}{\sqrt{2}}\right) \quad z_1=3cis\frac{\pi}{4}$

$Q=(0+4\cos 135°, 0+4\sin 135°)=\left(4\left(-\frac{1}{\sqrt{2}}\right),4\frac{1}{\sqrt{2}}\right)=\left(-2\sqrt{2},2\sqrt{2}\right)$

$P=A+Q-0=\left(\frac{3}{\sqrt{2}}-2\sqrt{2},\frac{3}{\sqrt{2}}+2\sqrt{2}\right)=\left(\frac{3-4}{\sqrt{2}},\frac{3+4}{\sqrt{2}}\right)$

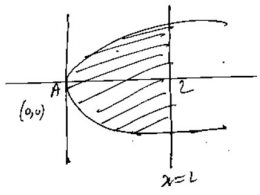
$=\left(\frac{-1}{\sqrt{2}},\frac{7}{\sqrt{2}}\right)=\frac{1}{\sqrt{2}}+i\frac{7}{\sqrt{2}}=3cis\frac{\pi}{4}\left(1+\frac{4i}{3}\right)=cis\left(\frac{\pi}{4}\right)(3+4i)=(3+4i)e^{i\frac{\pi}{4}}$

21. Let P(x, y) be any point on the curve C $\frac{dy}{dx}=\frac{1}{y} \qquad \int ydy=\int dx \qquad \frac{y^2}{2}=x+C$

$\uparrow M(2,2)\Rightarrow\frac{2^2}{2}=2+C\Rightarrow C=0 \qquad \therefore\frac{y^2}{2}=x\Rightarrow y^2=2x$

$x=2 \qquad y=\sqrt{2x}$

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$$Area = 2\int_0^2 \sqrt{2x}dx = 2\sqrt{2}\int_0^2 \sqrt{x}dx = 2\sqrt{2}\frac{2}{3}\left(x^{3/2}\right)_0^2$$

$$= \frac{4\sqrt{2}}{3}\left(2\sqrt{2}\right) = \frac{16}{3} = A9\,units \qquad \frac{3A}{2} = 8$$

22. $e^y = t \Rightarrow e^y\frac{dy}{dx} = \frac{dt}{dx}\frac{dt}{dx} - (2\sin x)t = -\sin x\cos^2 x$

$$I.F. = e^{2\cos x}\; e^y e^{2\cos x} = \int e^{2\cos x}\left(-\sin x\cos^2 x\right)dx$$

$$= \frac{1}{2}\cos^2 x e^{2\cos x} - \frac{1}{2}\cos x e^{2\cos x} + \frac{e^{2\cos x}}{4} + c$$

$$x = \frac{\pi}{2} \qquad y = 0 \qquad c = \frac{3}{4}$$

$$y = \log\left[\frac{\cos^2 x}{2} - \frac{\cos x}{2} + \frac{1}{4} + \frac{3}{4}e^{-2\cos x}\right]$$

$$Put\;\; x = 0\;\; y = \log\left(\frac{1}{4} + \frac{3}{4}e^{-2}\right)\;\; \alpha = \frac{1}{4} \qquad \beta = \frac{3}{4} \therefore 4(\alpha+\beta) = 4$$

23. $x = 10! \qquad y = 10c_1 9c_8 \qquad \frac{y}{9x} = 5$

24. $$= \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix}\begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \end{vmatrix} = (1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2 \therefore k = 1$$

25.

X	0	1	2	3
$P(X)$	$\frac{1}{12}$	$\frac{5}{12}$	$\frac{5}{12}$	$\frac{1}{12}$

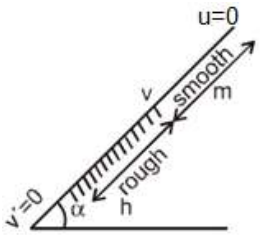
$$\sigma^2 = \frac{7}{12}$$

$$96\sigma^2 = 8\times 7 = 56$$

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PHYSICS

26.

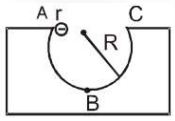


On smooth surface $a_1 = g\sin\alpha \therefore v^2 = u^2 + 2a_1 s_1 = 0 + 2g\sin\alpha.m$

on rough surface $a_2 = g\sin\alpha - \mu g\cos\alpha \therefore v'^2 = v^2 + 2a_2 s_2$

$$O = 2mg\sin\alpha + 2g(\sin\alpha - \mu\cos\alpha)n \Rightarrow \mu = \left[\frac{m+n}{n}\right]\tan\alpha$$

27. When cylinder reaches pt B
then block get shifted by x $\therefore$ but since than there is no ext.
force therefore com remain at its position

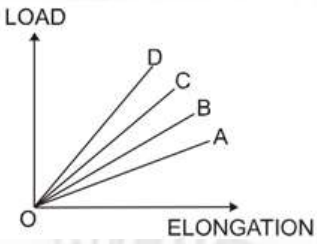


$$\left[(R-r)-x\right]m = Mx \therefore x = \frac{m(R-r)}{M+m}$$

28. $I = \mu L^2$ $\mu =$ reduce mass

29. $$R = vt = \sqrt{2gD}\sqrt{\frac{2(H-D)}{g}} = 2\sqrt{(H-D)D}$$

30.



$$\frac{F/A}{\Delta\ell/\ell} = Y \quad \frac{F}{\Delta\ell} = \frac{Y\pi r^2}{\ell} \Rightarrow \frac{F\ell}{Y\pi} \times \frac{1}{\Delta\ell} = r^2$$

$\Rightarrow Y$ & ℓ are same for all then

for same load, $r\alpha \dfrac{1}{\sqrt{\Delta\ell}}$

31. $$C = Cv + \frac{R}{1-\eta} \ldots (1)$$

$T^2V^{-3} =$ constant

$PV = nRT$

$P^2V^2V^{-3} =$ constant

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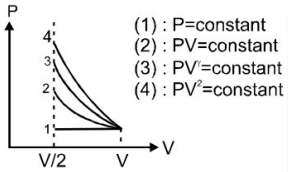
$P^2V^{-1} = \text{constant}$

$PV^{-1/2} = \text{constant}$

$N = -1/2$

$C = \frac{5R}{2} + \frac{2R}{3} = \frac{19R}{6}$

32.



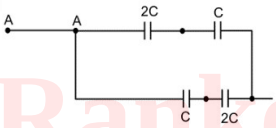
So, final pressure is maximum for PV^2 = constant.

33. Initially, $F = \frac{kq_1q_2}{r^2} \ldots\ldots\ldots\ldots (1)$

Finally, $4F = \frac{kq_1q_2}{16R^2} \ldots\ldots\ldots\ldots (2)$

$\Rightarrow \frac{4kq_1q_2}{r^2} = \frac{kq_1q_2}{16R^2}$ or $R = \frac{r}{8}$

34. Rearrange the circuit



$C_{eq} = \frac{2C}{3} \times 2$

35. $V_{ab} = L\frac{dl}{dt} + 1R \quad 8 = L \times 1 + 2 \times R \quad 4 = -L \times 1 + 2 \times R$

Solving the above equations

we get $R = 3\Omega \; L = 2H$

36. In photoelectric experiment, speed of fastest emitted electron is given by

$\frac{1}{2}mv_{\max}^2 = \frac{hc}{\lambda} - w$

Case-I: $\frac{1}{2}mv^2 = \frac{hc}{\lambda} - w \quad \ldots\ldots (1)$

Case-II: $\frac{1}{2}mv'^2 = \frac{hc}{3\lambda/4} - w \quad \frac{1}{2}mv'^2 = \frac{4hc}{3\lambda} - w \quad \ldots\ldots (2)$

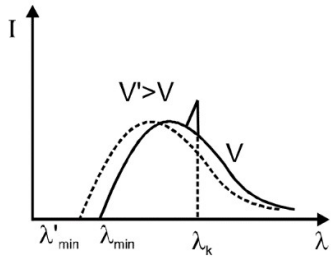
From equations (1) & (2) above,

$v'^2 = \frac{4}{3}v^2 + \frac{w}{3}$

Hence, $v' > v\sqrt{\frac{4}{3}}$

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37. By increasing the operating voltage, λ_k does not change but $\lambda_{\min}$ decreases.



Hence $\lambda_k - \lambda_{\min}$

38. For relaxed eye adjustment,

$m = \dfrac{f_0}{f_e}$ So, $f_e = 10cm$ and separation between lenses $= f_0 + f_e = 110cm$

39. $\dfrac{1}{f} = \dfrac{2(\mu-1)}{R}\dfrac{1}{12} = \dfrac{2(1.5-1)}{R} = \dfrac{1}{R}$

$R = 12cm$

$\dfrac{1}{f_w} = \dfrac{2[\mu_r - 1]}{R}\dfrac{1}{f_w} = \dfrac{2\left[\dfrac{1.5}{1.35}-1\right]}{12} = \dfrac{2\times 0.15}{12\times 1.35}$

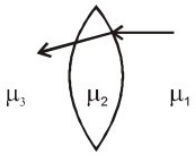
$\dfrac{1}{f_w} = \dfrac{1}{54}\dfrac{1}{f_{eq}} = \dfrac{2}{54} \Rightarrow f_{eq} = 27cm$

40. $[v] = [k]\left[\lambda^a \rho^b g^c\right] \Rightarrow LT^{-1} = L^a M^b L^{-3b} L^c T^{-2c}$

$\Rightarrow LT^{-1} = M^b L^{a-3b+c} T^{-2c} \Rightarrow a = \dfrac{1}{2}, b = 0, c = \dfrac{1}{2}$

So, $V^2 = kg\lambda$

41. (A)

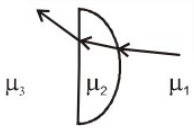


$\mu_2 = \mu_3$

As there is no deviation. As the light bends towards normal in denser medium $\mu_2 > \mu_1$

p – A & C

(B)



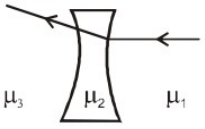
As light bends away from normal &

$\mu_2 < \mu_1$

$\mu_3 < \mu_2$

$q - B \& D$

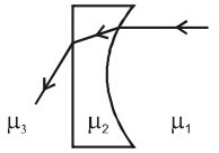
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(C) 

$\mu_2 = \mu_3$ (As no deviation)

$\mu_2 > \mu_1$ (As light bends + towards normal)

r – C & A

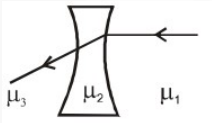
(D) 

$\mu_2 < \mu_1$

$\mu_3 < \mu_2$

As light bends away from normal

s – B, D

(D) 

$\mu_2 = \mu_3$ As no deviation of light

$\mu_2 < \mu_1$ As light bend away from normal

t – C & B

42. Nuclear density is independent of A

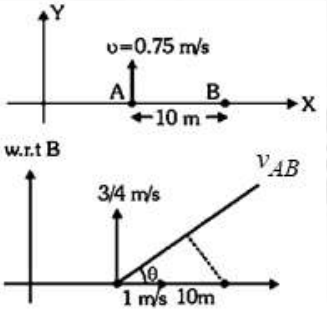
43. As constantan and manganin have very small values of temperature coefficient of resistance they are used in making standard resistance coils.

44. S1 is true but S2 is false as per the theory of p-n junction

45. $T = 2\pi\sqrt{\frac{1}{g}}$, $T\alpha\frac{1}{\sqrt{g}}$

on Everest, g decreases, so T increases, so it moves slowly.

46.



$$\tan\theta = \frac{3}{4}$$

$$t = \frac{10\cos\theta}{\sqrt{1+\frac{9}{16}}} = \frac{10\times\frac{4}{5}}{\frac{5}{4}} = \frac{160}{25} = 6.4\,\text{sec}.$$

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47. Given $v_B = 0.5\sqrt{gr}$

Assume block leave the contact at $C, N = 0$

$$\frac{mv_C^2}{r} = mg\cos\theta \quad .....(1)$$

from energy conservation $\frac{1}{2}mv_B^2 + mgr(1-\cos\theta) = \frac{1}{2}mv_C^2 .....(2)$

from equation (1) and (2)

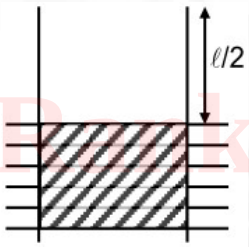
$$\frac{1}{2}m\left(\frac{1}{4}gr\right) + mgr(1-\cos\theta) = \frac{1}{2}mgr\cos\theta$$

$$\Rightarrow \cos\theta = \frac{3}{4} \Rightarrow \theta = \cos^{-1}\frac{3}{4}$$

48. $I = f \times \Delta t$ and $F = \dfrac{m\left(\sqrt{2gh_2} + \sqrt{2gh_1}\right)}{\Delta t}$

$$F = \frac{100\times10^{-3}\left(\sqrt{2\times9.8\times0.625} + \sqrt{2\times9.8\times2.5}\right)}{0.01} \Rightarrow F = 105\,N$$

49.



$$f^1 = \frac{v}{\frac{4\ell}{2}} = \frac{v}{2\ell} = 500Hz$$

50. $[36 - \rho_\ell v_1]g = [48 - \rho_\ell v_2]g$

$$\left[36 - \rho_\ell\left(\frac{36}{9}\right)\right]g = \left[48 - \rho_\ell\left(\frac{48}{\rho_0}\right)\right]g$$

solving

$\rho_0 = 3$

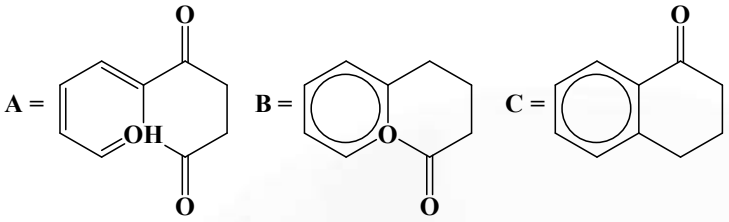
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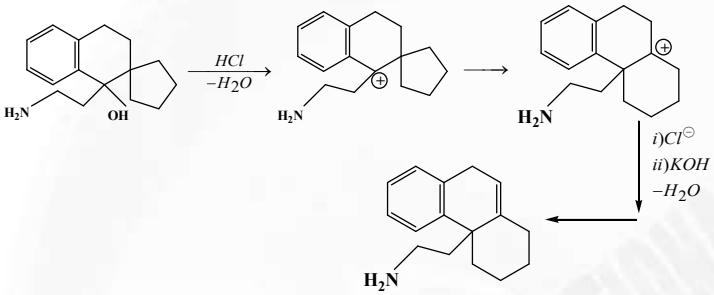
51. Cis-2-butene has more dipole moment than trans-2-butene

52.



53. Named reactions

54.



55.

$$CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-OCOCH_2-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_3 \xrightarrow{dil.H_2SO_4} \quad [A]$$

$$CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-OH + HOOC-CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_3$$

[C] [B]

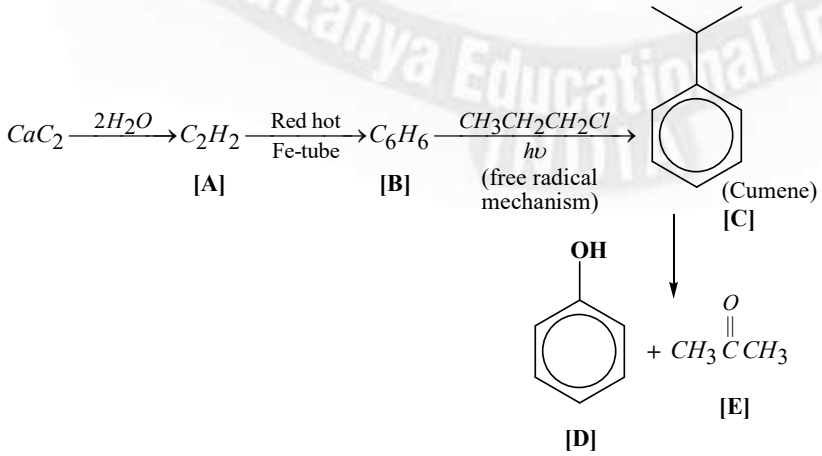
$$\xrightarrow{C_rO_3 / H^{\oplus}}$$

$$H_3C-CH_2-\underset{\underset{CH_3}{|}}{C}=O$$

Which is not 'B'

56. Fructose has $\alpha-$ hydroxy ketone group which Tautomerises to aldehyde group in presence of base. So it reduces Tollen's reagent.

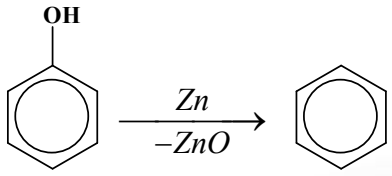
57. Conversions



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A) $C_2H_2 \xrightarrow[\text{reagent}]{\text{Tollen's}} Ag-C\equiv C-Ag\,(\text{White ppt})$

B)



C) Phenol is million times acidic than ethanol

D) $H_3C-\underset{\underset{CH_2}{|}}{C}=0+R-Mg\times \xrightarrow{H_3O^{\oplus}} H_3C-\overset{\overset{O}{|}}{C}-OH$

58. Complex is - $\left[Zn(NH_3)_4\right]^{2+}$

$CNO=4$

$Hyb-sp^3$. Shape is tetrahedral

59. * The complex $\left[Ni(en)_3\right]^{3+}$ is violet

* Primary valency is ionizable but secondary valency is non ionizable

60. 1) Angular momentum of electron $mvr=\frac{nh}{2\pi}$

2) Radial node for 4d orbital is $(n-\ell-1)=4-2-1=1$

3) Radius of nth orbit $r_n=\frac{0.529\times n^2}{2}A^0$

61. Phosphorous and Arsenic forms $d\pi-d\pi$ bond with transition elements only

62. Og – Noble Gas; Ts – 17th group; Lv – 16th group; Mc-15th group

63. XeO_2F_2

No. of L.P. = 1

Hyb - sp^3d

shape – Sea saw

64. Equilibrium is possible only in closed vessel

65. At 0K to 115 K the particles to move and oscillate.

Hence, entropy of system increase.

66. $Cr_2O_7^{2-}$ – orange colour due to charge transfer

MnO_4^{-2} – green colour due to d – d transition

67. $Sm^{2+}-4f^6$; $Pm^{2+}-4f^5$; $Nd^{2+}-4f^4$; $Eu^{3+}-4f^6$; $Gd^{3+}-4f^7$

68. Homoleptic complexes -

$\left[Fe(CN)_6\right]^{3-}, \left[Ni(CO)_4\right], [FeF_6]^{3-}, \left[Cr(H_2O)_6\right]^{3+}$

Diamagnetic is - $\left[Ni(CO)_4\right]$

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Paramagnetic are - $\left[Fe(CN)_6\right]^{3-}, \left[FeF_6\right]^{3-}, \left[Cr(H_2O)_6\right]^{3+}$

69. H_2O contain Intermolecular H-bond

70. A mix of ethanol and acetone shows positive deviation.

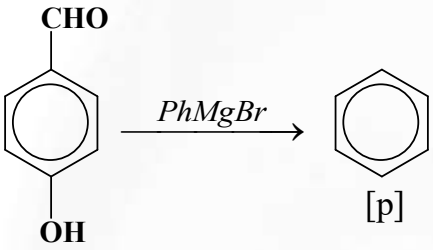
71. $CH_3CH_2NH_2 \xrightarrow{NaNO_2+HCl} H_3C\,CH_2N_2^{\oplus}Cl^{\ominus} \xrightarrow{H_2O} CH_3-CH_2-OH+N_2$

$45gm$

$?\ (4.5)gm$

$(or)45\times10^{-1}gm$

72.



[p]

73. $\lambda m°HCOOH = \lambda m°HCO\bar{O} + \lambda m°H^+$

$= 349.6 + 54.6$

$= 404.2\,S\,cm^2mole^{-1}$

$\alpha = \frac{\lambda m}{\lambda m°} = \frac{46.1}{404.2} = 0.11405 = 11.405\times10^{-2} = 11$

74. $t_{1/2} = \frac{0.693}{k} = \frac{0.693}{5.5\times10^{-14}} = 0.126\times10^{14} = 126\times10^{11}$

75. $m = \frac{1000\times M}{1000\times d - M\times Mwt \text{ of solute}}$

$= \frac{1000\times3}{1000\times1.25-3\times58.5}$

$= \frac{3000}{1250-175.5} = \frac{3000}{1074.5} = 2.79$

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